



PERTANDINGAN WORLDSKILLS MALAYSIA KATEGORI BELIA (WSMB) TAHUN 2026

(IT NETWORK SYSTEM ADMINISTRATION)

MODUL A

LINUX ENVIRONMENT

PERINGKAT AKHIR

MASA : 6 JAM

NAMA: _____

No. KP: _____

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Contents

Contents

Contents	2
Introduction to Test Project	3
Introduction	3
Description of project and tasks.....	4
Part 1 – wsmb2026.my domain	4
HQ-EDGE	4
CLIENT	4
INTSRV	5
Part 2 – itnsa.my domain	6
DC-EDGE.....	6
DC-SRV1	6
DC-SRV2 & DC-SRV3.....	7
Part 3 – Public Internet Domain.....	8
ISP-SRV.....	8
INET.....	8
PUB-CLNT	Error! Bookmark not defined.
Appendix 1: Network Table.....	9
Appendix 2: Topology	10
Appendix 3: wsmb2026.my Users	10

Introduction to Test Project

The following is a list of sections or information that must be included in all Test Project proposals that are submitted to WorldSkills Malaysia Belia (WSMB).

- Contents including list of all documents, drawings and photographs that make up the Test Project
- Introduction/overview
- Short description of project and tasks
- Instructions to the Competitor
- Marking scheme (incl. assessment criteria)
- Other

Introduction

The competition has a fixed start and finish time. You must decide how to best divide your time. Please **carefully read** the following instructions!

When the competition time ends, please leave your station in a running state. The assessment will be done in the state as it is. *No reboot will be initiated as well as powered off machines will not be powered on!*

Set all user password and credential with **Skills39** unless being specifically stated with password. Please use the information below for all the servers and clients.

System Configuration

Region/timezone: Asia/Kuala Lumpur

Locale: English US (UTF-8)

Key Map: English US

When tasked with configuring SSL or TLS you, use certificate from certificates server in the topology. If unable to do so, you can use a self-signed-certificate but you will lose some mark. Please take care to hide any warnings.

Software

For testing purpose, all linux hosts have been installed with the following test tools: **smbclient, curl, lynx, dnsutils, ldap-utils, ftp, lftp, wget, ssh, nfs-common, rsync, telnet, traceroute, tcptraceroute**

Files

- ITNSA-FINAL-WSMB 2026_ModuleA.docx
- debian-13.2.0-amd64-DLBD-1.iso

These files are available for all competitors that will used for various tasks along this Test Project

Description of project and tasks

You are the IT consultant responsible for WSMB2026 and you have been tasked with implementing a complex IT-environment.

Part 1 – wsmb2026.my domain

This part consists of Linux-based server and clients. Your task is to install and configure the required services. You can choose to reinstall or change the settings as you see fit. Refer to Appendix 1: Network Table for information about the subnets used.

HQ-EDGE

Functional testing will be applied on this server. Make sure to configure the hostname and IP address based on the topology on Appendix 1: Network Table

Routing

- Enable forwarding on this server to make this server act as a router
- Install and configure **Routing Services (frr)**
 - Enable routing protocol to receive routes on the Public Internet Network.
 - Use OSPF in a single backbone area as the dynamic protocol

Site-to-Site VPN

- Configure GRE over IPsec VPN between **itnsa.my** site and **wsmb2026.my** site
- Use certificate issued by WSMB2026-CA for authentication.
 - If certificate is unavailable, pre-shared key can be used with mark penalty.
- Use **tun1** as the interface name
 - Use 192.168.200.0/30 as the tunnel subnet

NFTables

- Input and forward this server should be **blocked by default**
- Traffic originating from **internal network** should be explicitly allowed by default
- For traffic related to DMZ network, only required traffic should be allowed
- This server should allow to ping any hosts
- Configure **PAT** for all hosts within wsmb2026.my domain

DHCP

- Install and configure **Dynamic Host Configuration Protocol (isc-dhcp-server)** on this server
 - IP range : 192.168.10.120 – 192.168.10.180
 - DNS Server : INTSRV server
 - Default Router : HQ-EDGE server
- Enable Dynamic DNS (DDNS) record update based on the assignment of IP from DHCP.

CLIENT

IP Address should be obtained through DHCP. This device is pre-installed with the GNOME interface. You can choose to reinstall or change the settings as you see fit. Make sure to configure the hostname based on the topology on Appendix 1: Network Table

Lightweight Directory Access Protocol (LDAP)

- Enable LDAP authentication against INTSRV
- Make sure LDAP user able to login to this server

Testing services

- Firefox will be used for testing services. Make sure Firefox is configured accordingly.
- Do not delete the emails that have been exchanged for testing.

INTSRV

Functional testing will be applied on this server. Make sure to configure the hostname and IP address based on the topology on Appendix 1: Network Table

Domain Name System (bind9)

- Install and configure **Domain Name System (bind9)** for wsmb2026.my.
- Configure internal **wsmb2026.my** name server
 - Create static A and PTR records for all servers of wsmb2026.my domain.
 - Create a service record for LDAP. Make sure it points to INTSRV.wsmb2026.my
 - The protocol is TCP, port is 389, priority is 10 and weight is 50
- Configure **itnsa.my** name server
 - Ensure this server is able to reach the authoritative server without replicating the entire zone.
 - This server should also contain the reverse zone for itnsa.my.
 - Make sure the zone is transferred through the network (AXFR/IXFR) and not manually transferred.
- Configure **"ISP-SRV.internet.com"** as root hint server. Remove other root hint records.
- Configure and enable DNS recursion.

Lightweight Directory Access Protocol (LDAP)

- Install and configure **LDAP (openldap)**
- Add user objects according to **Appendix 3: wsmb2026.my** users.
 - Never use plaintext password as user password.
- Create a group "cn=users,ou=groups,dc=wsmb2026,dc=my".
 - Ensure every user in LDAP is the member of this group.

Certificate Authority

- Configure Certificate Authority. The certificate should have subject as "C=MY, CN=WSMB2026-CA".
 - Make sure CA is available at **/etc/ssl/CA** directory
- All PKI usage should be based on the keys generated by this Certificate Authority

Samba

- Create a file server using Samba.
 - Create the user "smbuser" with password of "Skills39". You do not need to connect samba with LDAP.
 - Create the following shares, all files should exist within /data parent folder.
 - /public: everyone including guests should be able to read files. Only logged-in users are able to modify files
 - /internal: only logged-in users are able to access. Disable guest access to this share.

Part 2 – itnsa.my domain

DC-EDGE

Functional testing will be applied on this server. Make sure to configure the hostname and IP address based on the topology on Appendix 1: Network Table

Routing

- Enable forwarding on this server to make this server act as a router
- Install and configure **Routing Services (frr)**
 - Enable routing protocol to receive routes on the Public Internet Network.
 - Use OSPF in a single backbone area as the dynamic protocol

Site-to-Site VPN

- Configure GRE over IPsec VPN between **itnsa.my** site and **wsmb2026.my** site
- Use certificate issued by WSMB2026-CA for authentication.
 - If certificate is unavailable, pre-shared key can be used with mark penalty.
- Use **tun1** as the interface name
 - Use 192.168.200.0/30 as the tunnel subnet

NFTables

- Configure **NAT** for related services provided in the Server Farm Subnet accessible from Public Internet Network

DC-SRV1

Functional testing will be applied on this server. Make sure to configure the hostname and IP address based on the topology on Appendix 1: Network Table. This server will be pre-installed with Zeal Docs and Visual Studio Code.

Ansible

- Install and configure Ansible to automate tasks in DC-SRV3.
 - Ensure that DC-SRV1 is able to reach DC-SRV3 using Ansible. Host is defined using their hostname.
 - The task will be specified on “DC-SRV2 & DC-SRV3” section
 - **Note: Services within DC-SRV2 should be configured manually. Automation is only applicable on DC-SRV3. Functional testing will be applied on both servers.**

E-Mail

- Install and configure **e-mail server (postfix dovecot)**
- Use domain **mail.itnsa.my**
- Configure SMTPS and IMAPS service support SSL/TLS.
 - Use certificate generated by WSMB2026-CA
- All users should be able to freely exchange emails using Mail Service
 - Make sure users is able to login using credentials from LDAP. Use the mail attribute of LDAP user.
- Make sure all mail domains are able to exchange email freely

HAProxy

- Install and configure the **HAProxy** service.
 - Configure HAProxy to load balance “https://www.itnsa.my” between servers using round robin.
 - Configure HAProxy to load balance “https://intra.itnsa.my” between servers using round robin.
- Add Header “via-proxy” with the hostname of current HAProxy host
- This proxy should do TLS termination.
- External clients from Public Internet should be able to access web server through proxy
- Ensure that HTTP traffic are redirected to HTTPS traffic by default.

DC-SRV2 & DC-SRV3

Functional testing will be applied on this server. Make sure to configure the hostname and IP address based on the topology on Appendix 1: Network Table.

Web

- Install and configure **web service (apache)**
- As you expect a higher load in the future, the team has decided to use **Ansible** to configure the web servers.
 - Note: Services within DC-SRV2 should be configured manually. Automation is only applicable on DC-SRV3. Functional testing will be applied on both servers.
 - Note: For grading, the experts will restore the initial state of DC-SRV3, and apply the ansible configuration on DC-SRV3 by running the command
 - **"ansible-playbook /opt/ansible/configure-web-srv3.yml" on DC-SRV1**
 - Note: Make sure that the script will run with this exact command as changes are not allowed
- TLS termination will be done at reverse proxy, make sure web listen to port 80.
- Configure default website **http://www.itnsa.my** site
 - Make sure the content is as below with the "[hostname]" set as the server's hostname.
 - **"<h1><center>Welcome to website of [hostname] </center></h1>"**
- Configure website **http://intra.itnsa.my** site
 - When client access this directory, there should be basic authentication from wsmb2026.my LDAP.
 - Display the hostname only when authentication succeeded.

Domain Name System (bind9)

- Install and configure **Domain Name System (bind9)** for itnsa.my.
- As you expect a higher load in the future, the team has decided to use **Ansible** to configure the web servers.
 - Note: Services within DC-SRV2 should be configured manually. Automation is only applicable on DC-SRV3. Functional testing will be applied on both servers.
 - Note: For grading, the experts will restore the initial state of DC-SRV3, and apply the ansible configuration on DC-SRV3 by running the command
 - **"ansible-playbook /opt/ansible/configure-dns-srv3.yml" on DC-SRV1**
 - Make sure that the script will run with this exact command as experts are not allowed to adapt/change
- Configure internal primary zone of **itnsa.my** name server on DC-SRV2.
 - Create static A and PTR records for all servers of itnsa.my domain.
 - Create CNAME and MX records required by services
- Configure internal secondary zone of **itnsa.my** name server on DC-SRV3.
 - The forward and reverse zones should be transferred via DNS zone transfer.
 - Manual transfer of zone is not allowed.
- Configure external **itnsa.my** name server at DC-SRV2
 - Create static A, CNAME, and MX records external records of itnsa.my domain.
- Make sure all server only answers queries for its authoritative DNS zones.

Part 3 – Public Internet Domain

This part consists of Linux based server and clients to simulate an internet environment. Your task is to install and configure the required services. All of the IP addressing in this server is preconfigured. You can choose to reinstall or change the settings as you see fit. Refer to the "Network Table" appendix for information about the subnets used.

ISP-SRV

This server is pre-configured with IP addressing and hostname

Domain Name System (DNS)

- Install and configure **internet.com** name server
 - Create static A, CNAME, and MX records for all servers of **public internet domain**.
 - Create appropriate delegations to resolve DNS records for itnsa.my.

Web Server

- Install and configure any relevant web server.
 - Create a website with URL **http://www.internet.com**
 - Create HTML with website showing content
 - `"<center><h1>Hello from Internet</h1></center>"`

E-mail

- Install and configure **e-mail server (postfix dovecot)**
 - Configure SMTP and IMAP service with domain mail.internet.com
 - Create two users: user1 and user2 with password Skills39 for email testing
 - Make sure all mail domains are able to exchange email freely

INET

This server is pre-configured with IP addressing and hostname

Routing

- Enable forwarding on this server to make this server act as a router
- Install and configure **Routing Services (frr)**
 - Enable routing protocol to receive routes.
 - Use OSPF in a single backbone area as the dynamic protocol

DHCP

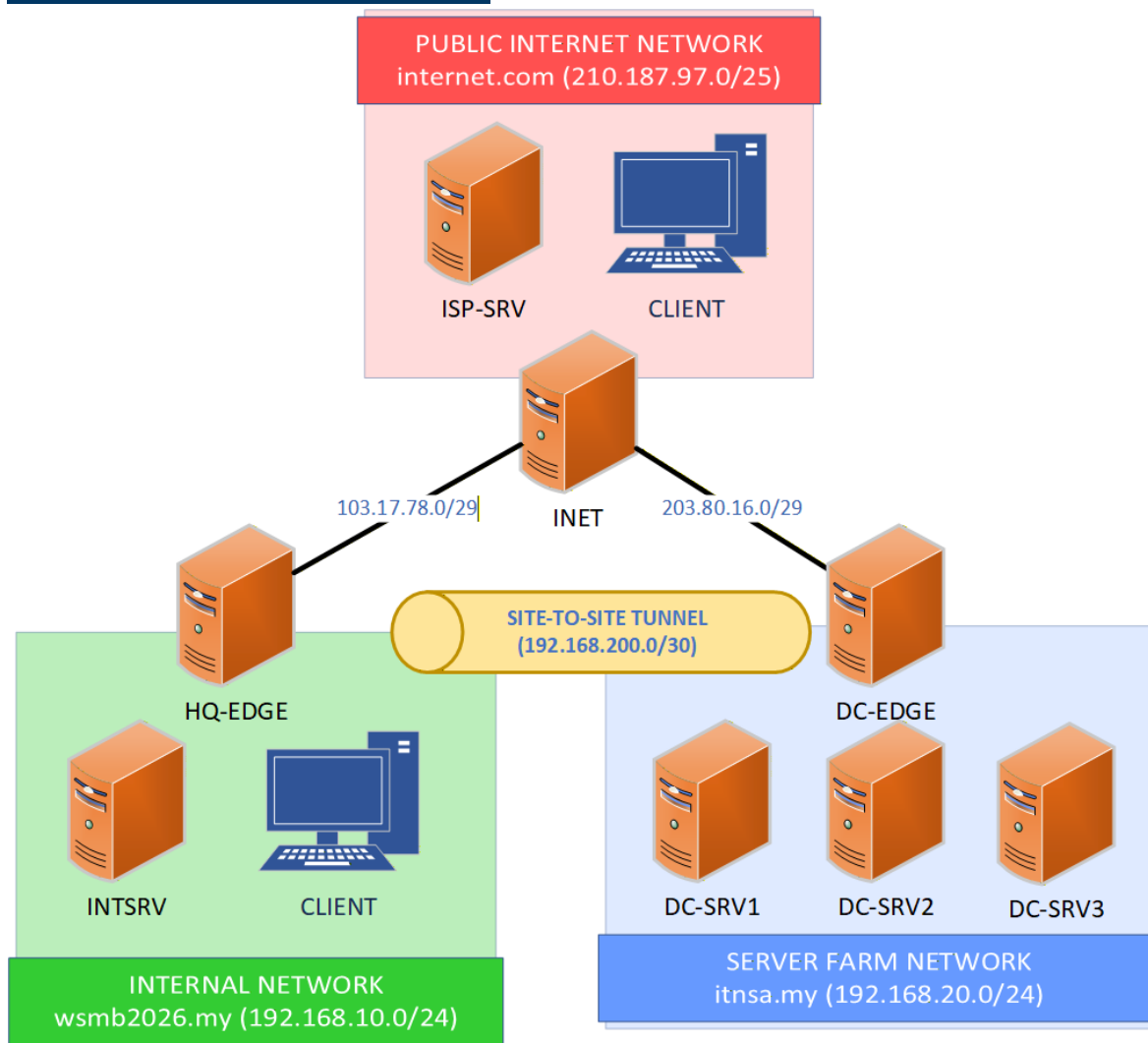
- Install and configure **Dynamic Host Configuration Protocol (isc-dhcp-server)** on this server
 - IP range : 210.187.97.60 – 210.187.97.80
 - DNS Server : ISP-SRV server
 - Default Router : INET server

Appendix 1: Network Table

Hostname	Operation System	Domain	IP Address(es)
INTSRV	Debian 13.2 CLI	wsmb2026.my	192.168.10.10/24
CLIENT	Debian 13.2 (GNOME)	wsmb2026.my	DHCP Assignment
HQ-EDGE	Debian 13.2 CLI	wsmb2026.my	192.168.10.254/24 (Internal) 103.17.78.1/29 (HQ-EDGE <> INET)
DC-SRV1	Debian 13.2 (GNOME)	itnsa.my	192.168.20.11/24
DC-SRV2	Debian 13.2 CLI	itnsa.my	192.168.20.12/24
DC-SRV3	Debian 13.2 CLI	itnsa.my	192.168.20.13/24
DC-EDGE	Debian 13.2 CLI	itnsa.my	192.168.20.254 (Server Farm) 203.80.16.1/29 (DC-EDGE <> INET)
INET	Debian 13.2 CLI	internet.com	103.17.78.6/29 (HQ-EDGE <> INET) 203.80.16.6/29 (DC-EDGE <> INET) 210.187.97.126/25 (Internet)
ISP-SRV	Debian 13.2 CLI	internet.com	210.187.97.10/25
CLIENT	Debian 13.2 (GNOME)	-	DHCP Assignment

Network	CIDR
wsmb2026.my - Internal	192.168.10.0/24
Itnsa.my - Server Farm	192.168.20.0 /24
Internet.com – Public Internet	210.187.97.0/25
wsmb2026.my <> ISP Network	103.17.78.0/29
itnsa.my <> ISP Network	203.80.16.0/29
WSMB <> ITNSA Site-to-Site	192.168.200.0/30

Appendix 2: Topology



Appendix 3: wsmb2026.my Users

Full Name	Username(uid)	Password	Email address
Jimmy Choo	jimmy	Skills@39	jimmy.choo@itnsa.my
Samad Said	samad	Skills@39	samad.said@itnsa.my
Ananda Krishnan	krishnan	Skills@39	ananda.krishnan@itnsa.my